Smart Home Environmental Monitoring System

🏠 Project Overview

This project implements a **Smart Home Environmental Monitoring System** using a **two-microcontroller architecture**. The system measures environmental parameters such as:

- 🌡️ Temperature
- 💧 Humidity
- 🌫️ Air Quality (MQ-2 Gas Sensor)
- 💡 Ambient Light (Photoresistor / LDR)

The collected data is processed and transmitted from a **Sensor Node** to an **Output Node**, where it is displayed on a **16×2 I^2C LCD**.

⚙️ System Architecture

```
[ Sensor Node – Arduino Nano 33 BLE Sense Rev2 ]
   ├── LDR (Light Sensor) → A0
   ├── MQ-2 Gas Sensor → RC Filter → Op-Amp → A1
   ├── HS3003 Temp/Humidity (on-board)
   └── UART TX → Data Transmission

                  ↓ UART (Serial1)

[ Output Node – Arduino MKR1010 ]
   ├── UART RX → Data Reception
   └── I²C LCD (16×2 PCF8574) → Display
```

🔌 Hardware Components

◇ Sensor Node (Arduino Nano 33 BLE Sense Rev2)

- Photoresistor (LDR)
- MQ-2 Gas Sensor (Flying Fish module)
- LM358 Op-Amp
- Resistors:
  - 10 kΩ (LDR divider)
  - 10 kΩ (RC filter + amplifier)
- Capacitors:
  - 0.1 µF (RC filter)
  - 0.1 µF (decoupling capacitor)

◇ Output Node (Arduino MKR1010)

- 16×2 LCD with I^2C backpack (PCF8574)

🔧 Circuit Design

🧠 LDR Voltage Divider (Light Sensor)

```
3.3V
  |
 [ LDR ]
  |
  ├──→ A0
  |
 [ 10kΩ ]
  |
 GND
```

🌫 MQ-2 Signal Conditioning

Step 1: RC Low-Pass Filter

```
MQ-2 AOUT → 10kΩ → ● → Op-Amp (+)
                        |
                      0.1µF
                        |
                       GND
```

Step 2: LM358 Amplifier (Gain ≈ 2)

```
Gain = 1 + (Rf / Rin) = 2

Rf = 10kΩ
Rin = 10kΩ
```

Decoupling Capacitor:

```
Pin 8 (VCC) ──||── Pin 4 (GND)
        0.1 µF
```

🔌 Communication (UART)

Sensor Node (Nano BLE)	Output Node (MKR1010)
D1 (TX1)	Pin 13 (RX1)
GND	GND

- Baud Rate: **9600**
- Protocol: **CSV**

Example transmitted packet:

```
65.2,18.3,24.50,42.10
```

📟 Display Interface (I^2C LCD)

LCD Pin	MKR1010
VCC	3.3V
GND	GND

SDA	Pin 11
SCL	Pin 12

Software Description

Sensor Node Responsibilities

- Reads analog data from LDR and MQ-2
- Applies signal conditioning
- Reads temperature & humidity (HS3003)
- Sends formatted data over UART

Output Node Responsibilities

- Receives UART data
- Parses CSV values
- Displays data on LCD

Code Structure

```
/Sensor_Node/
   ├── LDR_read.ino
   ├── gas_sensor_read.ino
   ├── temp_humidity.ino
   └── sensorData_transmission.ino

/Output_Node/
   └── receiveData_and_LCD_display.ino
```

Key Design Features

ADC & Signal Conditioning

- LDR uses voltage divider for analog measurement
- MQ-2 uses RC filter + op-amp for:
    - noise reduction

    - signal amplification

☑ Embedded Communication

- UART (Serial1) used for inter-device communication
- CSV protocol ensures easy parsing

☑ Human-Readable Interface

- LCD displays:
    - Light %
    - Gas %
    - Temperature (°C)
    - Humidity (%RH)

📊 Example Output

Serial Output:

```
LDR: 62.5% | Gas: 18.2% | Temp: 24.7°C | Humidity: 45.3%RH
```

LCD Output:

```
L:62% G:18%
T:24.7C H:45%
```

🔬 Testing & Validation

- Verified ADC readings using Serial Monitor
- Tested gas detection using lighter (controlled exposure)
- Verified light response using flashlight
- Confirmed UART transmission reliability
- Validated LCD display updates in real-time

⚠ Important Notes

- MQ-2 must be powered with **5V**
- Common ground is required between both Arduino boards
- LCD may require I^2C address change (`0x27` or `0x3F`)
- Op-amp must be powered at **3.3V** for ADC compatibility

🚀 Future Improvements

- Add **wireless communication (BLE/WiFi)**
- Implement **interrupt-based sampling**
- Add **power-saving modes**
- Implement **automation (e.g., LED alerts)**
- Add **data logging / cloud dashboard**

👨‍💻 Author Contribution

- Sensor circuit design (LDR, MQ-2, filtering, amplification)
- Embedded firmware for sensor acquisition
- UART communication implementation
- LCD display interface design
- System integration and debugging

📚 References

- Arduino Nano 33 BLE Sense Rev2 Documentation
- Arduino MKR1010 Documentation
- LM358 Datasheet
- MQ-2 Gas Sensor Datasheet
- Arduino_HS300x Library

⭐ How to Run the Project

1. Upload sensor code to Nano 33 BLE Sense Rev2
2. Upload receiver code to MKR1010
3. Connect UART (TX → RX) and GND
4. Power MQ-2 using 5V
5. Connect LCD via I^2C
6. Open Serial Monitor (optional)
7. Observe real-time environmental data

🎯 Summary

This project demonstrates a complete embedded system combining:

- analog sensing
- signal conditioning
- embedded communication
- real-time display

It reflects core concepts of **IoT systems, embedded design, and smart environments**.